

Keeping Up With the Rest of the World: Digitalised Note-Taking and Paper-Solving Among English-Medium Students in Bangladesh and the Effects on Their Cognitive Control

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Abstract

This study examines the transition from traditional paper-based studying to digital note-taking and past-paper solving among O-Level and A-Level English-medium students in Bangladesh. As international examination boards increasingly adopt digitised assessment formats, it is essential to assess the effects of this transition on students' physical strain and cognitive focus to inform effective educational practices. Employing a cross-sectional, correlational design, survey data were collected from current candidates ($N = 85$), who were classified into Predominantly Digital ($n = 57$) and Predominantly Paper ($n = 28$) study groups. Composite scores for Eye Strain (ESS), Hand Strain (HSS), and Cognitive Control Index (CCFI) were analysed using paired and independent samples t-tests, as well as Pearson correlation coefficients. The results showed that musculoskeletal strain ($X = 6.32$) exceeded ocular strain ($X = 5.86$) across the cohort, $t(84) = 1.838$, $p = 0.0696$, although this difference did not reach statistical significance at $\eta^2 = 0.05$. Additionally, no statistically significant differences in cognitive control scores were found between digital ($X = 16.25$) and paper users ($X = 16.93$), $t(83) = -0.870$, $p = 0.3868$. Daily study duration was not significantly correlated with physical or cognitive strain ($p > 0.05$), suggesting that fatigue patterns are determined primarily by medium-specific ergonomic factors rather than study duration. These findings support the adoption of a balanced, hybrid approach, emphasising ergonomic awareness as educational institutions transition to e-learning environments.

Keywords: Digitalisation, Cognitive Control, English-Medium Education, Eye Strain, Musculoskeletal Strain, Past-Paper Preparation, Ergonomics, Bangladesh.

1. Introduction

As education becomes increasingly digitalised, academic institutions worldwide are adopting laptops, tablets, and digital styluses for instruction, note-taking, and assessment. With major international testing bodies such as Cambridge Assessment International Education (CAIE) preparing for full digital exams, secondary and higher secondary students in developing nations are under pressure to meet these global standards.

Traditionally, academic preparation in Bangladesh's English-medium curriculum relied on analogue methods, such as handwriting lecture summaries and completing bound volumes of past examination papers. In the past decade, portable devices, stylus-based digital paper apps, and online learning platforms have become increasingly popular. Digital media offers flexibility by enabling users to combine typing,

